

CLASS 9 · ICSE · SELINA CONCISE MATHEMATICS · DPP SHEET

Chapter 25: Complementary Angles

Daily practice problems: 10 questions with an answer key

Try every question on your own first. The answer key is on the last page.

1. Evaluate $\frac{\cos 37^\circ}{\sin 53^\circ}$.
2. Evaluate $\operatorname{cosec} 31^\circ - \sec 59^\circ$.
3. Evaluate $\frac{\tan 26^\circ}{\cot 64^\circ} + \frac{\sin 18^\circ}{\cos 72^\circ}$.
4. Evaluate $\cos^2 40^\circ + \cos^2 50^\circ$.
5. Evaluate $\tan 5^\circ \tan 25^\circ \tan 30^\circ \tan 65^\circ \tan 85^\circ$.
6. Express $\cos 78^\circ + \cot 62^\circ$ in terms of angles between 0° and 45° .
7. If $\sin 2A = \cos(A + 15^\circ)$, where $2A$ is acute, find A .
8. If $\tan 3\theta = \cot(\theta + 10^\circ)$, where 3θ is acute, find θ .
9. Prove that $\sin 63^\circ \cos 27^\circ + \cos 63^\circ \sin 27^\circ = 1$.
10. If A, B, C are the angles of a triangle, prove that $\tan \frac{A+B}{2} = \cot \frac{C}{2}$.

Answer Key

1. 1

2. 0

3. 2

4. 1

5. $\frac{1}{\sqrt{3}}$

6. $\sin 12^\circ + \tan 28^\circ$

7. 25°

8. 20°

9. It equals $\sin^2 63^\circ + \cos^2 63^\circ = 1$

10. $\frac{A+B}{2} = 90^\circ - \frac{C}{2}$, and $\tan \left(90^\circ - \frac{C}{2} \right) = \cot \frac{C}{2}$